

Dulara Madhusanka

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PROFESSIONAL SUMMARY

Data Science undergraduate and AI Research Assistant specializing in curriculum learning, graph neural networks, deep learning experimentation, and statistical analysis. Experienced in developing reproducible PyTorch research pipelines, designing entropy-based training strategies, evaluating models under low-resource and noisy conditions, and communicating technical work through peer-reviewed publications and conference workshops. First author of two papers accepted for publication at MERCon 2026 and the IJCAI-ECAI 2026 GlobalSouthAI Workshop, and co-author of a paper presented at the Global South ML Workshop at ICML 2026.

EDUCATION

BSc (Hons) in Information Technology – Specializing in Data Science

2024–2028

Sri Lanka Institute of Information Technology (SLIIT)

Current GPA: **3.6 / 4.0**

- **Mathematical Foundations:** Probability and Data Analysis, Linear Algebra, Discrete Mathematics, Statistics
- **Computer Science Core:** Data Structures and Algorithms, Database Management Systems, Systems Engineering
- **Professional Development:** Certified in *Introduction to Data Analytics in Google Cloud* (March 2025)

RESEARCH EXPERIENCE

Undergraduate Research Assistant – AI Systems Research, SLIIT

Nov 2025–Present

- Built scalable graph neural network training pipelines for curriculum-learning and spatio-temporal modeling research.
- Developed entropy-based difficulty scoring and feature-masking strategies to improve training stability across low-resource datasets.
- Automated controlled experiments to track GPU utilization, runtime efficiency, validation performance, and model behavior across curriculum schedules.
- Contributed to three peer-reviewed research papers and supported the SLIIT Research Grant, *A Brain-Inspired Human Behavior Understanding Framework using Curriculum Learning*.

Independent Mathematics Instructor

Mar 2025–Present

- Taught probability, combinations, discrete mathematics, and statistics to 900+ students through structured problem-solving sessions.
- Translated advanced mathematical ideas into intuitive frameworks that strengthened analytical and quantitative reasoning.

RESEARCH PUBLICATIONS

MERCon 2026

Accepted; publication pending

“Tag Entropy: A Structure-Agnostic Curriculum Learning Framework for Graph Neural Networks” **First Author**

- Proposed an architecture- and structure-agnostic graph difficulty measure using normalized Shannon entropy for curriculum learning with GNNs.

IJCAI-ECAI 2026 Workshop on GlobalSouthAI

Accepted; publication pending

“Beyond Easy-to-Hard: Investigating Tag-Entropy-Based Training Order in Graph Neural Networks” **First Author**

- Investigated easy-to-hard and hard-to-easy curriculum orderings across multiple GNN architectures and graph-classification datasets.

Global South ML Workshop at ICML 2026

Presented, Seoul, South Korea – July 2026

“Confusion-Aware Transfer Teacher Curriculum Learning Framework: Disentangling Scoring and Pacing Effects”

Co-Author

- Co-authored a deep learning optimization framework that disentangles curriculum scoring, pacing, and feature-separation effects.
- **Preprint:** [arXiv:2606.17706](https://arxiv.org/abs/2606.17706)

CONFERENCE SPEAKING

Moratuwa Engineering Research Conference (MERCon 2026)

2026

Speaker – Does Order Matter? Curriculum Learning for Training Deep Neural Networks

- Presented curriculum-learning concepts for GNN training, including graph difficulty estimation and the effects of training order.
- Discussed practical easy-to-hard and hard-to-easy pacing strategies developed through ongoing graph-learning research.

6th International Conference on Advanced Research in Computing (ICARC 2026) *Online – Feb 2026*

Guest Speaker – Technical Workshop on Curriculum Learning

- Delivered a one-hour technical workshop on GNN curriculum-learning architectures and a hands-on session on pacing strategies.

SELECTED PROJECTS

Entropy-Based Curriculum Training Pipeline

[GitHub](#)

- Developed a GPU-accelerated curriculum-learning pipeline with entropy-driven difficulty scoring and dynamic masking.
- Achieved a 2.84% validation accuracy improvement and a 39% F1-score improvement under noisy training conditions.

Conditional Diffusion Model for Controlled Image Generation

[GitHub](#)

- Developed a conditional diffusion architecture in PyTorch with classifier-free guidance and inference validation safeguards.
- Added adjustable class-conditioned sampling and safeguards against checkpoint and weight-initialization failures.

RFM Customer Segmentation and Lifetime Value Analytics

[GitHub](#)

- Built an end-to-end analytics pipeline using Python, DuckDB SQL, Power BI, statistical hypothesis testing, and custom DAX measures.
- Validated segment behavior using Kruskal–Wallis, Chi-square, and Gini-index analyses and designed an interactive customer-value dashboard.

Validated Monte Carlo Risk Engine

[Live App](#) – [GitHub](#)

- Developed a Student's t -distribution risk simulator with rolling backtests and Kupiec POF validation, supporting 10,000+ simulations.
- Optimized memory use for deployment on a constrained 512 MB cloud instance.

RAG-Based Intelligent Booking Ecosystem

[GitHub](#)

- Developed a FastAPI and ChromaDB retrieval-augmented platform with vector embeddings and modular semantic-retrieval endpoints.

Jarvis – Offline Voice-Controlled AI Assistant

[GitHub](#)

- Built a fully local assistant using Python, speech-recognition pipelines, and Llama 3.2 through Ollama for low-latency offline interaction.
- Resolved acoustic echo cancellation feedback and integrated Linux system-level automation for local application and process control.

TECHNICAL SKILLS

Programming & Query Languages: Python, SQL, Java, Bash, R

Machine Learning & Research: PyTorch, TensorFlow, GNNs, CNNs, curriculum learning, diffusion models, statistical testing

Data Analytics & BI: Power BI, DuckDB, DAX, RFM segmentation, hypothesis testing

MLOps & Engineering: FastAPI, Docker, Git, Linux (Fedora), Streamlit, Render, MongoDB

Generative AI: Llama 3.2 (Ollama), LangChain, LangGraph, ChromaDB, RAG pipelines

Quantitative Methods: Monte Carlo simulation, VaR/CVaR, Student's t -distribution, Kupiec POF backtesting

Languages: English (Professional), Sinhala (Native)

REFEREES

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